

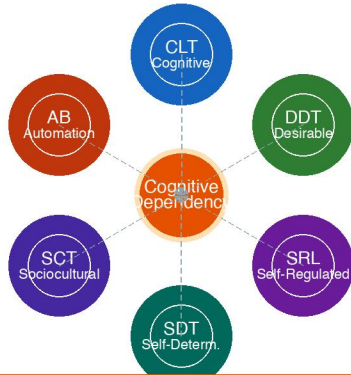
The Cognitive Dependency Framework

Theoretical Convergence and Meta-Analytic Evidence for AI-Induced Metacognitive Compromise

Meta-Analysis: k = 65 • 381 effect sizes • N = 8,247 • Nov 2022 – Jan 2026

A. Theoretical Convergence

Six frameworks predict attenuated metacognitive effects



A Priori Hypothesis (H3)

Metacognitive outcomes will show smaller effects than other dimensions, reflecting cognitive dependency

B. Meta-Analytic Evidence

Three-level random effects model with robust variance estimation



Critical Finding

Metacognitive outcomes: $g = 0.28 [-0.24, 0.80]$
Confidence interval crosses zero $\square$ non-significant
Pattern consistent with cognitive dependency hypothesis
GRADE certainty: Very Low (serious imprecision)

C. Evidence Synthesis

- ☒ Cognitive & affective: consistent benefits
- ☒ GenAI enhances immediate performance
- ☐ Behavioral: limited evidence (k=16)
- ☐ Metacognitive: non-significant effect
- ☐ Independent self-regulation not developed

Selected Moderators

Bloom's Lower-Order	$g = 0.81^{***}$
Bloom's Higher-Order	$g = 0.32$ (ns)
STEM Disciplines	$g = 0.71^{***}$
Humanities/Social	$g = 0.48^*$

Practical Implication

Design AI interventions with explicit metacognitive scaffolding and fading protocols to promote internalization

D. AI-Induced Metacognitive Compromise (AIMC) Framework

A developmental model linking theoretical mechanisms to observed patterns

Level 1: AI-Assisted Metacognition

- External regulation via AI prompts
- Scaffolded monitoring & planning
- Performance with support

Strong evidence ($g = 0.64$)

Level 2: Meta-AI Awareness

- Recognition of AI limitations
- Strategic tool selection
- Calibrated trust in outputs

Mixed evidence

Level 3: Independent Metacognition

- Autonomous self-regulation
- Internalized strategies
- Transfer to novel contexts

Weak evidence ($g = 0.28$, ns)

Evidence Gap

Current interventions may not bridge Level 1 $\square$ Level 3

Theoretical Convergence

Six established frameworks predict metacognitive vulnerability

Empirical Confirmation

Meta-analysis reveals significant differential effects

Design Imperative

AI scaffolding must include explicit fading protocols